DPP No: 40

Maximum Time 50 Min

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(COMPETISHUN CRASH COURSE)

PHYSICS

TARGET JEE-MAINS

SYLLABUS : WAVE OPTICS

1. Two coherent sources of intensities I_1 and I_2 produce an interference pattern. The maximum intensity in the interference pattern will be :–

(A) $I_1 + I_2$ (B) $I_1^2 + I_2^2$ (C) $(I_1 + I_2)^2$ (D) $(\sqrt{I_1} + \sqrt{I_2})^2$

2. Two wave are represented by the equations $y_1 = a \sin \omega t$ and $y_2 = a \cos \omega t$. The first wave :–

(A) leads the second by π (B) lags the seconds by π

(C) leads the second by $\frac{\pi}{2}$ (D) lags the seconds by $\frac{\pi}{2}$

3. The resultant amplitude of a vibrating particle by the superposition of the two waves

$y_1 = a\sin\left[\omega t + \frac{\pi}{3}\right]$ and $y_2 = a \sin \omega t$ is :–

(A) a (B) $\sqrt{2}$ a (C) 2a (D) $\sqrt{3}$ a

4. The energy in the phenomenon of interference :–

(A) is conserved, gets redistributed (B) is equal at every point

(C) is destroyed in regions of dark fringes (D) is created at the place of bright fringes

5. The phase difference corresponding to path difference of x is :–

(A) $\frac{2\pi x}{\lambda}$ (B) $\frac{2\pi\lambda}{x}$ (C) $\frac{\pi x}{\lambda}$ (D) $\frac{\pi\lambda}{x}$

6. Ratio of intensities of two light waves is given by 4 : 1. The ratio of the amplitudes of the waves is :

(A) 2 : 1 (B) 1 : 2 (C) 4 : 1 (D) 1 : 4

7. Two coherent monochromatic light beams of intensities I and 4I are superposed; the maximum and minimum possible intensities in the resulting beam are :

(A) 5I and I (B) 5I and 3I (C) 9I and I (D) 9I and 3I

8. An electromagnetic wave travelling through a transparent medium is given by $E_x(y,t) = E_{ox} \sin 2\pi\left[\frac{y}{5\times10^{-7}} - 3\times10^{14}t\right]$ in SI units. Then what is the refractive index of the medium?

(A) 1 (B) 2 (C) 3 (D) 4

9. If the ratio of the intensity of two coherent sources is 4 then the visibility $[(I_{max} - I_{min})/(I_{max} + I_{min})]$ of the fringes is

(A) 4 (B) 4/5 (C) 3/5 (D) 9

10. If the distance between the first maxima anBd fifth minima of a double slit pattern is 7 mm and the slits are separated by 0.15 mm with the screen 50 cm from the slits, then wavelength of the light used is

(A) 600 nm (B) 525 nm (C) 467 nm (D) 420 nm

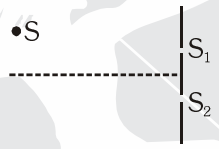
11. In Young's double slit experiment, the separation between the slits is halved and the distance between the slits and the screen is doubled. The fringe width is :–

(A) unchanged (B) halved (C) doubled (D) quadrupled

12. In Young's double slit experiment using sodium light (λ = 5898Å), 92 fringes are seen. If given colour (λ = 5461Å) is used, how many fringes will be seen

(A) 62 (B) 67 (C) 85 (D) 99

13. In YDSE, the source placed symmetrically with respect to the slit is now moved parallel to the plane of the slits it is closer to the upper slit, as shown. Then ,



(A) the fringe width will increase and fringe pattern will shift down.

(B) the fringe width will remain same but fringe pattern will shift up.

(C) the fringe width will decrease and fringe pattern will shift down.

(D) the fringe width will remain same but fringe pattern will shift down.

14. The contrast in the fringes in any interference pattern depends on :

(A) Fringe width (B) Wavelength

(C) Intensity ratio of the sources (D) Distance between the sources

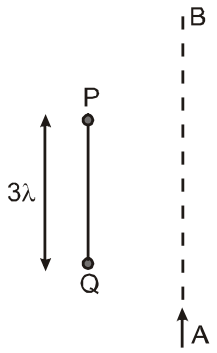
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15. Yellow light emitted by sodium lamp in Young's double slit experiment is replaced by monochromatic blue light of the same intensity :

(A) fringe width will decrease. (B) fringe width will increase.

(C) fringe width will remain unchanged. (D) fringes will become less intense.

16. Two coherent light sources each of wavelength λ are separated by a distance 3λ. The total number of minima formed on line AB which runs from $-\infty$ to $+\infty$ is:

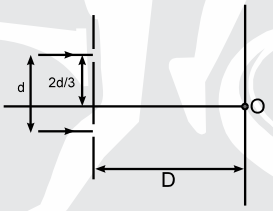


(A) 2 (B) 4 (C) 6 (D) 8

17. Two slits separated by a distance of 1 mm, are illuminated with red light of wavelength 6.5×10^{-7} m. The interference fringes are observed on a screen placed 1 m from the slits. Find the distance between the third dark fringe and the fifth bright fringe on the same side of the central maxima.

(A) 2.615 mm (B) 3.225 mm (C) 1.625 mm (D) 0.525 mm

18. In the figure shown if a parallel beam of white light is incident on the plane of the slits then the distance of the nearest white spot on the screen from O is: [assume d << D, λ << d]



(A) 0 (B) d/2 (C) d/3 (D) d/6

19. The Young's double slit experiment is performed with blue and with green light of wavelengths 4360 Å and 5460 Å respectively. If X is the distance of 4th maximum from the central one, then:

(A) X(blue) = X(green) (B) X(blue) > X(green)

(C) X(blue) < X(green) (D) $\frac{X(blue)}{X(green)} = \frac{5460}{4360}$

20. White light is used to illuminate the two silts in a Young's double slit experiment. The separation between the slits is b and the screen is at a distance d (> > b) from the slits. At a point on the screen directly in front of one of the slits, certain wavelengths are missing. Some of these missing wavelengths are :

(A) $\lambda = \frac{b^2}{d}$ (B) $\lambda = \frac{2b^2}{d}$ (C) $\lambda = \frac{2b^2}{3d}$ (D) None of these

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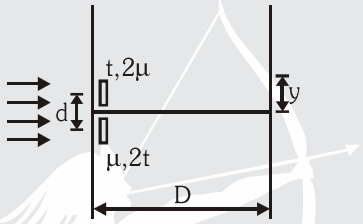
21. A source emitting two light waves of wavelengths 580 nm and 700 nm is used in a young's double slit interference experiment. The separation between the slits is 0.20 mm and the interference is observed on a screen placed at 150 cm from the slits. Find the linear separation between the first maximum (next to the central maximum) corresponding to the two wavelengths.

(A) 0.5 mm (B) 1.9 mm (C) 0.9 mm (D) 2.9 mm

22. In Young's double slit experiment, a mica sheet of thickness t and refractive index μ is introduced in the path of ray from the first source S_1. By how much distance the fringe pattern will be displaced

(A) $\frac{d}{D}(\mu-1)t$ (B) $\frac{D}{d}(\mu-1)t$ (C) $\frac{d}{(\mu-1)D}$ (D) $\frac{D}{d}(\mu-1)$

23. In the YDSE shown the two slits are covered with thin sheets having thickness t & 2t and refractive index 2μ and μ. Find the position (y) of central maxima



(A) zero (B) $\frac{tD}{d}$ (C) $-\frac{tD}{d}$ (D) None of these

24. A two slit Young's interference experiment is done with monochromatic light of wavelength 6000 Å. The slits are 2 mm apart. The fringes are observed on a screen placed 10 cm away from the slits. Now a transparent plate of thickness 0.5 mm is placed in front of one of the slits and it is found that the interference pattern shifts by 5 mm. The refractive index of the transparent plate is :

(A) 1.2 (B) 0.6 (C) 2.4 (D) 1.5

25. In a YDSE both slits produce equal intensities on the screen. A 100 % transparent thin film is placed in front of one of the slits. Now the intensity of the geometrical centre of system on the screen becomes 75 % of the previous intensity. The wavelength of the light is 6000Å and μ_{glass} = 1.5. The thickness of the film cannot be:

(A) 0.2 μm (B) 1.0 μm (C) 1.4 μm (D) 1.6 μm

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ANSWER KEY

1.	(D)	**2.**	(D)	**3.**	(D)	**4.**	(A)	**5.**	(A)
6.	(A)	**7.**	(C)	**8.**	(B)	**9.**	(B)	**10.**	(A)
11.	(D)	**12.**	(D)	**13.**	(D)	**14.**	(C)	**15.**	(A)
16.	(C)	**17.**	(C)	**18.**	(D)	**19.**	(C)	**20.**	(A)
21.	(C)	**22.**	(B)	**23.**	(B)	**24.**	(A)	**25.**	(D)